



Alleviating labeled data scarcity: a lightweight semi-supervised network for Moso bamboo age determination

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ABSTRACT

Accurate determination of Moso bamboo (*Phyllostachys edulis*) age is a critical task for efficient and sustainable bamboo forest management. However, existing methods face significant challenges: traditional manual assessment is subjective and labor-intensive, while advanced technologies like LiDAR are prohibitively expensive for widespread application. Furthermore, high-performance deep learning models, which offer a promising alternative, typically rely on large-scale labeled datasets, a resource that is particularly scarce and costly to acquire in the field of Moso bamboo. To address these limitations, we propose a lightweight, semi-supervised framework, the Dual-Color-Texture Moso Bamboo Age Decoupling Network (DCT-MBADNet). Our framework first leverages the Segment Anything Model (SAM) to isolate the bamboo culm, effectively eliminating complex background interference. A novel dual-stream feature decoupling module is then introduced to independently extract color degradation and texture evolution features, which are biologically significant indicators of bamboo age. A dynamic gating mechanism is employed to adaptively fuse these features. Simultaneously, we integrate an age-dependent dynamic threshold strategy within a Mean Teacher semi-supervised framework to synergistically utilize a small set of labeled data and a large volume of unlabeled data, thereby enhancing pseudo-label quality and model generalization. Experimental results demonstrate that our semi-supervised DCT-MBADNet achieves a test set accuracy of 89.6%, representing a 4.5% improvement over its fully supervised baseline. With a minimal parameter count of just 1.6 M, the proposed model provides a low-cost, robust, and deployable solution for precise Moso bamboo management and offers a novel paradigm for plant phenotyping analysis under data-scarce conditions.

1. Introduction

Moso bamboo is a cornerstone of China's forestry economy, renowned for its rapid growth, high yield, and a wide array of industrial applications [1,2]. Its unique biological characteristic of producing new culms from a subterranean rhizome system result in a distinct uneven-aged forest structure [3]. A well-established body of research demonstrates a critical link between a Moso bamboo stand's age composition and its overall biomass and productivity. Specifically, studies show that forests under intensive management (IM), which utilize precise age determination, achieve a 100% homogeneity score (via chi-square test) in culm age distribution, compared to only 41.7% homogeneity in extensively managed (EM) forests. This difference directly impacts stand stability and biomass yield [4–6]. Consequently, the

ability to accurately determine culm age is a foundational requirement for implementing effective management strategies and maximizing both ecological benefits and economic returns.

However, existing methods for bamboo age determination are fraught with significant limitations. Traditional empirical approaches, which rely on farmers' subjective visual assessments of culm sheath and color, lack consistency and reliability. This is further exacerbated by shifts in rural demographics and the gradual loss of inherited knowledge. Conversely, advanced technologies like Light Detection and Ranging (LiDAR) offer high precision for age classification [7,8], but their prohibitive equipment costs, operational complexity, and demanding data processing workflows have severely restricted their practical adoption at the grassroots level and in large-scale forest management. These shortcomings underscore a pressing need for a new

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method that is both low-cost and highly accurate, with the potential for widespread application.

In recent years, deep learning technology has shown immense potential in forestry image analysis. By accurately identifying plant phenotypic feature patterns and building a mapping relationship between feature space and target parameters, it provides a new approach for Moso bamboo age determination [9–12]. Age-related changes in Moso bamboo manifest as significant phenotypic biological characteristics: the culm color transitions from bluish-green to yellowish-brown [7,13], and the skin texture gradually roughens due to changes in cellulose content [14,15]. These color and texture changes provide a solid basis for feature learning with convolutional neural networks [16]. However, most existing models rely on fully supervised learning, which requires a large number of labeled samples to ensure model generalization. In the bamboo age determination task, obtaining age labels that match bamboo images is extremely costly, as it necessitates long-term tracking and tagging from the emergence of the bamboo shoot. This makes fully supervised learning methods difficult to popularize.

Semi-Supervised Learning (SSL) provides a new direction to alleviate the problem of insufficient labeled data by leveraging a small amount of labeled data and a large amount of unlabeled data. Typical methods like MixMatch [17], FixMatch [18], and FlexMatch [19] have shown performance close to fully supervised methods in general image classification tasks, demonstrating great potential under limited labeling

conditions. However, these methods often use a fixed threshold to filter pseudo-labels, which has obvious limitations in tasks with gradual phenotypic features. For Moso bamboo, 2-3 year-old and 4-5 year-old culms have similar colors and textures, forming transitional categories. A fixed threshold strategy can either over-filter these difficult-to-distinguish samples or introduce incorrect pseudo-labels, leading to decreased performance and reduced overall robustness in key categories. Additionally, complex background scenes in bamboo forests can interfere with feature extraction, and existing high-performance networks have many parameters, making them difficult to deploy efficiently in the field on resource-constrained devices.

To overcome these challenges, we propose a novel, lightweight semi-supervised framework based on a combined color-texture representation, which we term DCT-MBADNet. The primary contributions of this study are threefold:

- (1) Dual-stream Feature Decoupling Modeling: We designed a dual-stream decoupling architecture for color degradation and texture evolution, which aligns with the biological characteristics of Moso bamboo and effectively enhances the model's ability to represent age-related features.
- (2) Adaptive Dynamic Threshold Mechanism: We propose a dynamic threshold method that incorporates biological prior knowledge to

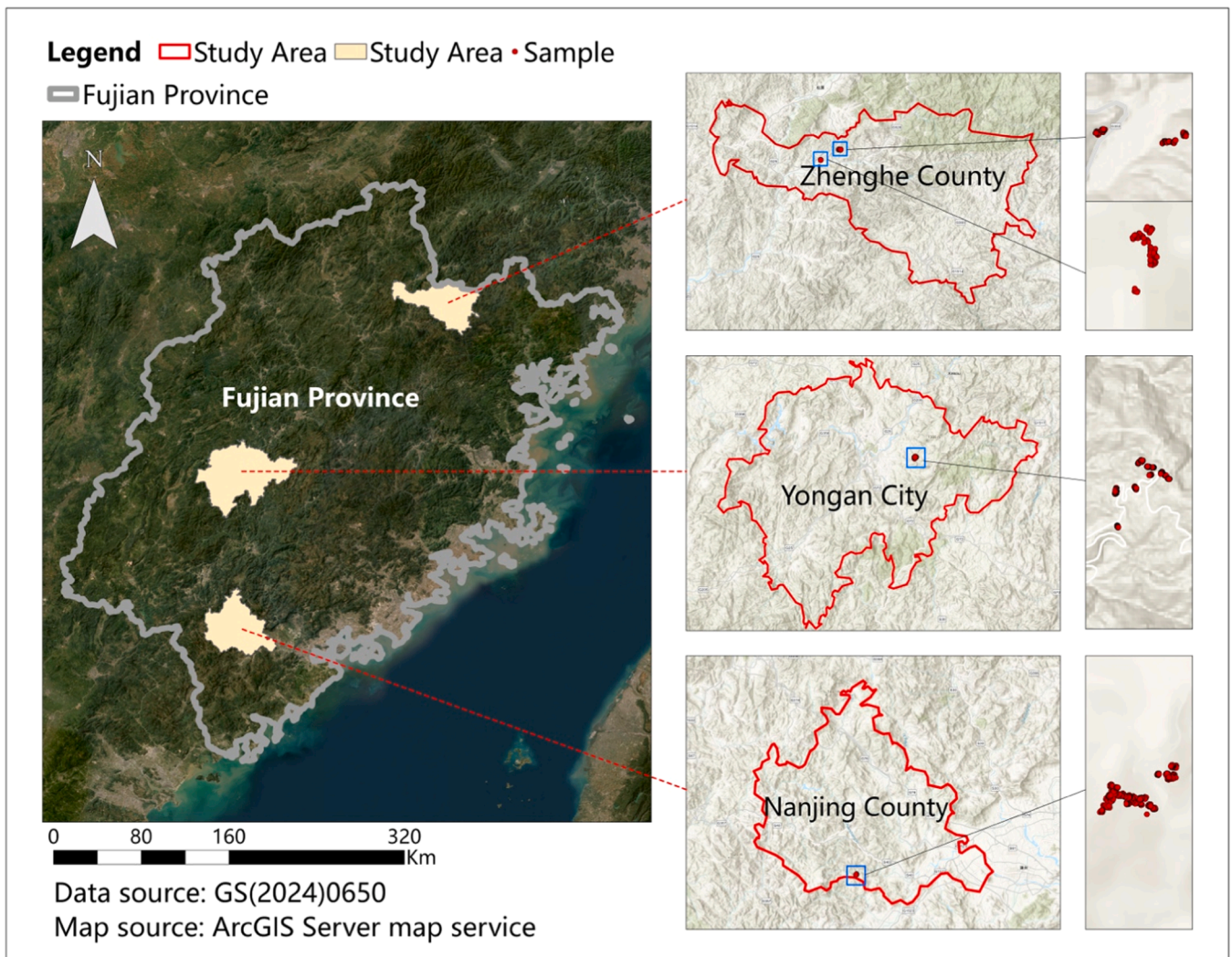


Fig. 1. Study area and sampling locations in Fujian Province, China. Red circles indicate field sampling sites within the three selected regions. Administrative boundaries are shown in grey.

adaptively filter high-confidence pseudo-labels, significantly mitigating the ambiguity in transitional category discrimination.

- (3) **Lightweight and Practical Value:** We constructed a lightweight semi-supervised model with only 1.6 M parameters that achieves high-precision classification even with limited labeled data and has the potential for field deployment, providing a low-cost, scalable solution for precise Moso bamboo management and ecological governance.

2. Dataset construction

2.1. Study area

As shown in Fig. 1 the study area spans from Nianshan in Zhenghe County in the north of Fujian Province (26.32°N, 118.85°E), Shangping Township in Yong'an City in the central part (25.97°N, 117.36°E) to Yongfeng Forest Farm in Nanjing County in the south (24.51°N, 117.35°E), covering three typical geographical units: the mid-mountain landscape in northern Fujian, the low-mountain and hilly area in central Fujian, and the red-soil tableland in southern Fujian. This region has a subtropical maritime monsoon climate, with an average annual temperature of 18.5–20.3°C and an annual precipitation of 1600–2200 mm. It features the simultaneous occurrence of rain and heat and distinct four seasons, providing highly suitable climatic conditions for the growth of Moso bamboo. As an important major Moso bamboo production area in China, the study area has a high vegetation coverage rate, with a bamboo forest area of 45,770 hectares. It forms a typical subtropical bamboo forest ecosystem with Moso bamboo as the dominant species, which has important ecological service functions and regional economic value.

2.2. Data collection

To ensure the diversity and generalizability of model data, as well as the consistency and accuracy of data collection during the research, forest stands with consistent forest stand structure and internal characteristics within the study area were selected for data acquisition. Commercially mainstream mobile phone models were adopted as data collection devices, including iPhone, Honor, OPPO, and others. During data collection, the ISO was fixed at 125, with strict attention to maintaining camera focus and avoiding overexposure to guarantee the quality of the captured data.

In China, bamboo age is commonly expressed in "degrees." Moso bamboo is classified into four age categories based on a widely accepted system: I-degree (1 year old), II-degree (2–3 years old), III-degree (4–5 years old), and IV-degree (6 or more years old). Data collection was specifically managed to maintain a balanced ratio of 1:1:1:1 across these four categories, thereby preventing a skewed class distribution from impacting model training and accuracy.

Labeled images were collected between July 12 and July 22, 2024. The true age of the bamboo was verified using pre-existing tags on the culms. After an initial cleaning process to remove poor-quality images, a total of 2,058 labeled images were retained, with a near-equal distribution: 528 for I-degree, 492 for II-degree, 513 for III-degree, and 525 for IV-degree and older.

Unlabeled images were collected from July 25 to August 14, 2024. These images were captured from sites intentionally separate from those used for the labeled dataset to prevent data leakage. During collection, experienced bamboo experts made a rough age assessment to ensure an approximate 1:1:1:1 ratio for each age group. A total of 5,396 unlabeled images were collected after cleaning.

The final dataset was partitioned to support the semi-supervised learning approach. The 2,058 labeled images were divided into a training set (1,647 images), a validation set (205 images), and a test set (206 images) using an 8:1:1 ratio. The 5,396 unlabeled images were used exclusively for semi-supervised training to enhance the model's

generalization capabilities, as shown in Fig. 2.

3. Methodology

3.1. Framework design

Based on the biological correlation between Moso bamboo phenotype and age, this study proposes a dual-stream semi-supervised learning framework for Moso bamboo age determination, DCT-MBADNet. Its core idea is to efficiently model color degradation and texture evolution features through a feature decoupling and curriculum learning strategy. The framework consists of three key stages, as shown in Fig. 3.

First, the main body of Moso bamboo culms is segmented based on SAM to eliminate the interference of complex backgrounds. The integrity of the mask is optimized through morphological post-processing to avoid the loss of effective features. Then, the dual-stream feature decoupling module constructed in this study is used to extract the color information and texture information of Moso bamboo respectively. The dynamic gating fusion module is employed to assign weights to the dual-stream feature information, preventing the feature information from being mixed and thus unable to be effectively utilized. Finally, the Mean Teacher framework is adopted, combined with the age-dependent dynamic threshold mechanism designed in this study to improve the quality of pseudo-labels. The supervised data and unsupervised data are used for the collaborative training of DCT-MBADNet, enhancing the generalization ability of the model.

3.2. Image pre-processing

To mitigate the effects of lighting variations, adaptive histogram equalization is applied to the masked images before encoding in the color stream branch. Furthermore, Moso bamboo images often contain background noise, such as the ground, other leaves and branches, and bright sky regions, which can interfere with the extraction of effective features. This study uses SAM [20] (Vit-H version) to generate initial masks for background removal. For the segmentation strategy, we employ SAM's 'SamAutomaticMaskGenerator' mode. This approach allows for zero-shot segmentation by automatically generating a dense grid of point prompts, thereby eliminating the need for any manual interaction or labeled inputs. To ensure high-quality segmentation and research reproducibility, the key parameters for the 'SamAutomaticMaskGenerator' were set as follows: 'points_per_side' is 1, 'pred_iou_thresh' is 0.91, 'stability_score_thresh' is 0.92, segment result as shown in Fig. 4.

The largest connected component is then retained to eliminate small noise areas, morphological closing operations are used to repair potential internal holes in the bamboo culm, and finally, Gaussian filtering is applied to smooth jagged edges, resulting in a fully preserved bamboo culm region. The average time consumption of this stage is 127 ms on our equipment. After obtaining the segmentation mask, texture priors are extracted in grayscale at the input end for subsequent multi-scale modeling in the texture stream.

3.3. Quantitative validation of the phenotypic traits of Moso bamboo

To rigorously quantify the color degradation and texture evolution characteristics of Moso bamboo and to provide biological justification and empirical support for the feature-prior design of the proposed dual-stream decoupling network, 200 epidermal images from culms of different age classes were randomly selected for phenotypic feature extraction. The quantitative descriptors included the mean RGB channel intensities, the G/R spectral ratio, and the gray-level co-occurrence matrix (GLCM) metrics of contrast and energy. The statistical results are summarized in Table 1.

The age-related changes in Moso bamboo are primarily manifested

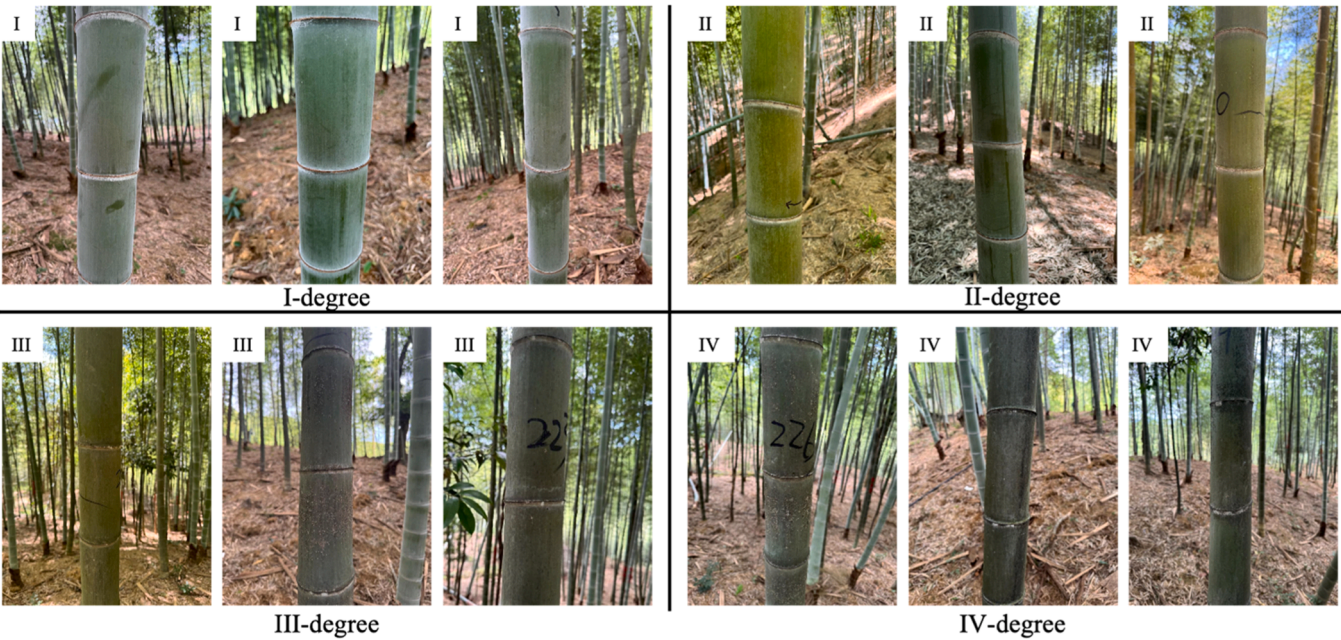


Fig. 2. Labeled Moso bamboo images.

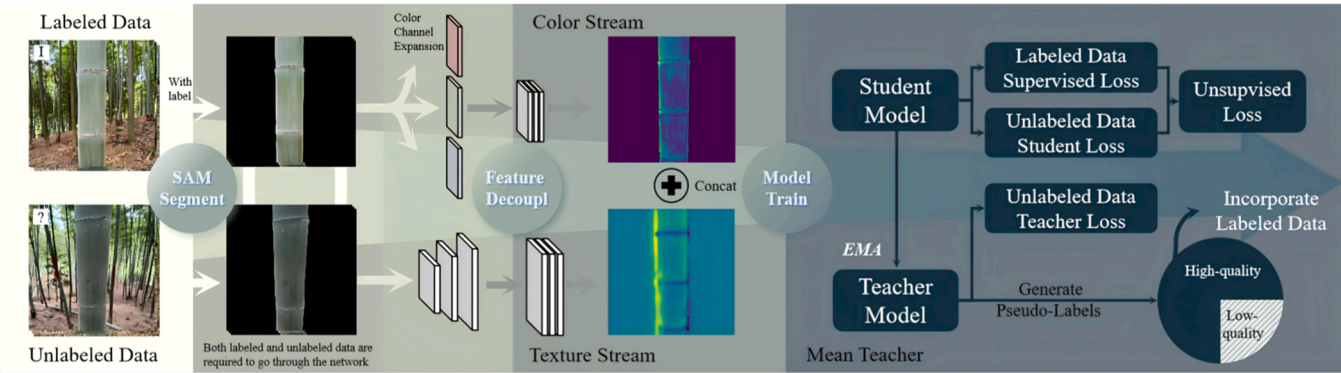


Fig. 3. Flowchart of DCT-MBADNet framework.

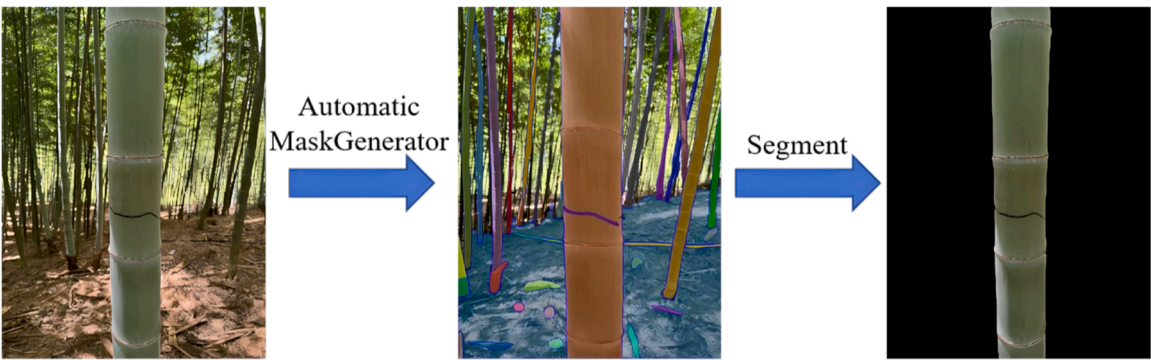


Fig. 4. Bamboo culm segmentation of Moso bamboo based on SAM.

through two dimensions: color degradation and texture evolution. Color degradation represents the direct spectroscopic manifestation of continuous chlorophyll breakdown, during which the culm color gradually shifts from dark green in one-year-old bamboos to bluish-green and eventually to a yellowish-brown tone. As shown in Table 1, the G/R ratio (an indicator highly sensitive to green chromaticity) reaches

its peak value of 1.108 in II-degree bamboo, corresponding well to the biologically observed verdant appearance at this stage, and then declines rapidly thereafter. This trend establishes a clear quantitative linkage between color traits and bamboo age. Meanwhile, texture evolution reflects the increased microstructural heterogeneity of the culm epidermis driven by natural weathering,

Table 1

Quantitative statistics of key phenotypic traits of Moso bamboo at different age stages.

Age	R_Mean	G_Mean	B_Mean	G/R Ratio	GLCM Contrast	GLCM Energy
I-degree	95.519	103.602	93.118	1.087	76.637	0.1706
II-degree	89.625	98.743	69.339	1.108	100.674	0.1697
III-degree	75.474	74.478	52.958	1.027	131.586	0.1664
IV-degree	72.697	73.682	61.22	1.022	162.423	0.1635

microbial colonization, and lignin accumulation. Quantitative analysis reveals a distinct monotonic relationship between texture and bamboo age: the GLCM contrast rises steadily from 76.637 in I-degree culms to 162.423 in IV-degree, strongly supporting the progressive roughening of the epidermal surface with age. Correspondingly, the GLCM energy decreases from 0.1706 to 0.1635, confirming the gradual loss of microstructural uniformity.

Therefore, the dual-stream decoupling architecture developed in this study is specifically designed to independently capture and adaptively fuse these two biologically meaningful modalities—color degradation and texture evolution—that effectively encode the aging process of Moso bamboo in visible-light imagery.

3.4. Core network architecture

The core of the proposed DCT-MBADNet is the dual-stream decoupling concept for color degradation and texture evolution. It is architecturally modeled based on the prior biological knowledge that Moso bamboo's color and texture features change with age. The overall network can be divided into three main modules: color stream feature encoding, multi-scale texture extraction, and dynamic gating fusion. The overall architecture is shown in Fig. 5.

The construction of the Color Degradation Feature Extraction Module is based on the regular changes in the color of the Moso bamboo culm surface with age. This study first expands RGB images to a 6-channel multi-dimensional color representation, including the original RGB channels, a brightness-hue channel, and a red-green color difference channel, through a color space transformation module to capture age-related color change patterns of Moso bamboo more comprehensively.

To further clarify the feature construction process, the brightness-hue and red-green color difference channels are computed as follows:

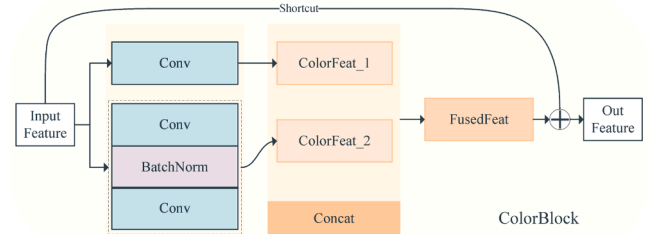
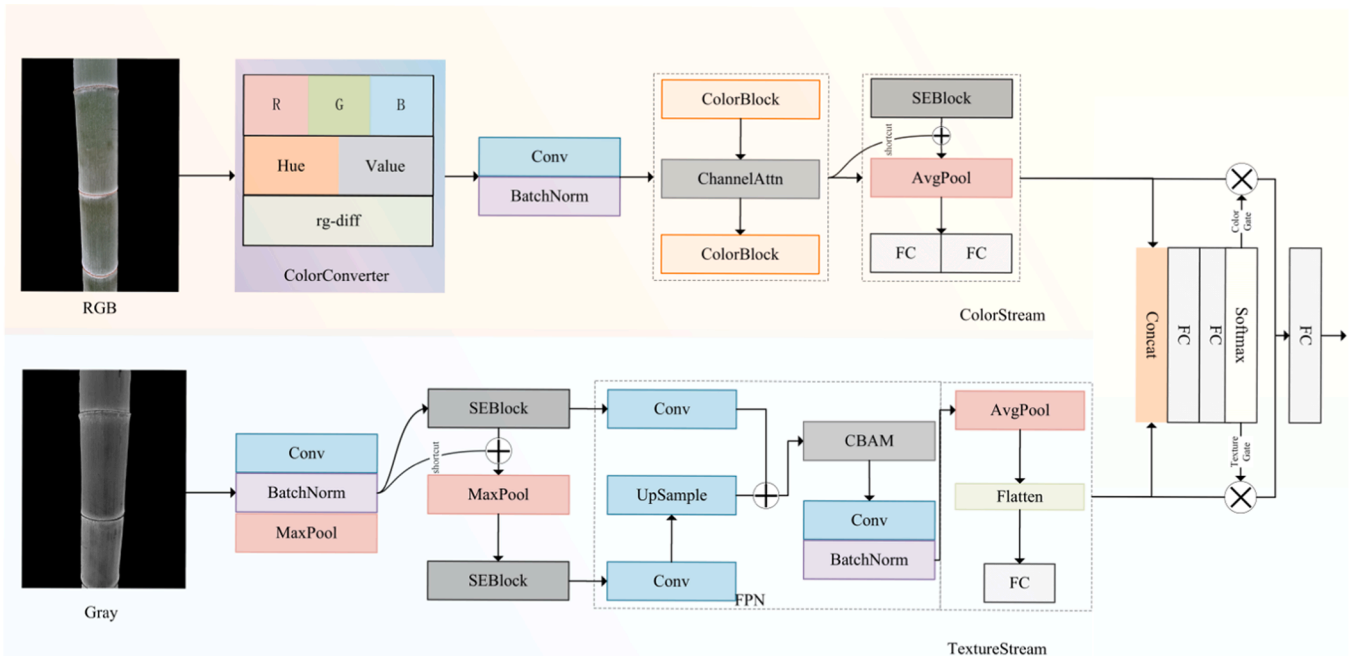
$$BHC(x,y) = \frac{1}{2} \left(\frac{R(x,y) + G(x,y) + B(x,y)}{3} + H(x,y) \right) \quad (1)$$

$$RGdiff(x,y) = R(x,y) - G(x,y) \quad (2)$$

Where $R(x,y)$, $G(x,y)$, $B(x,y)$ represent the pixel intensities of the RGB channels at spatial position (x,y) , and $H(x,y)$ is the hue component obtained from the HSV color space. Formula(1) constructs a brightness-hue composite channel that integrates overall luminance and chromatic tendency, emphasizing gradual color fading of bamboo culms. Formula(2) captures the red-green chromatic contrast, which reflects pigment loss during bamboo aging and has proven effective for distinguishing younger and older culms.

Next, a multi-scale ColorBlock is used to capture both local and regional color patterns, enabling precise recognition of the gradual color features of bamboo culms while maintaining computational efficiency. Its structure is shown in Fig. 6. Additionally, the SE attention mechanism [21] is introduced, which uses global pooling and channel importance learning to enable the model to adaptively focus on the most critical color channels for age discrimination, significantly enhancing its ability to recognize color degradation patterns on the Moso bamboo surface.

The evolution of Moso bamboo skin texture exhibits significant multi-scale characteristics: young bamboo has a smooth and delicate

**Fig. 6.** The structure of Color Block.**Fig. 5.** The network structure of DCT-MBADNet.

surface dominated by dense micro-textures, while with increasing age, the skin gradually roughens, and older bamboo forms macro-texture structures like distinct longitudinal fissures. To accurately capture this multi-scale texture evolution, this study designed a lightweight Feature Pyramid Network (FPN) [22]. This module uses two cascaded lightweight residual SE blocks to extract multi-scale features: high-resolution feature layers capture the fine micro-textures of young bamboo, while low-resolution feature layers identify the macro-texture structures of older bamboo. The two-layer FPN architecture enables a bottom-up fusion of the feature pyramid, effectively integrating texture information from different scales. Finally, combined with the CBAM [23] mechanism, it adaptively enhances the most discriminative texture features for age classification, accurately extracting the evolution pattern of Moso bamboo skin texture from delicate to rough. Considering the complementarity of color degradation and texture evolution features in Moso bamboo age determination, this study designed a lightweight intelligent feature fusion module. It dynamically learns the importance weights of color and texture features through a simplified gating mechanism, with the formula (3)

$$f_{\text{fusion}} = \theta f_{\text{color}} + (1 - \theta) f_{\text{texture}} \quad (3)$$

where θ is dynamically learned through pooling layers and fully connected layers, enabling adaptive weighted fusion of the two feature types. Compared to traditional feature concatenation methods, this fusion strategy can automatically adjust the fusion weights based on the feature distribution of different samples, better balancing the contributions of color and texture information.

3.5. Loss functions and training strategy

The training framework adopts the Mean Teacher semi-supervised learning paradigm, whose core idea is the prediction consistency between a Teacher Model and a Student Model, leveraging a small amount of labeled data and a large amount of unlabeled data to improve model performance. The parameters of the teacher model are obtained by smoothly updating the student model's parameters, thereby providing more stable predictions during training.

As Moso bamboo's phenotypic traits evolve with age, those in the transitional phase often exhibit similar characteristics, complicating accurate age determination due to subtle, hard-to-detect changes. Hence, selecting a loss function aligned with the biological nuances of Moso bamboo is crucial to enhance the model's ability to discern these subtle features, thereby improving age classification accuracy. This study employs Focal Loss [24] as the supervisory loss function, addressing the challenge of class difficulty imbalance in target detection tasks. By assigning varying weights to samples based on classification difficulty, the model focuses more on challenging samples. The specific formula(4) is:

$$L_{\text{sup}}(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t) \quad (4)$$

where p_t denotes the predicted probability assigned to the true class, representing the model's confidence in correct classification. The coefficient α_t is a balancing factor that controls the relative importance of different classes and compensates for the unequal distribution of easy and hard samples. The term $(1 - p_t)^\gamma$ acts as a focusing factor, down-weighting well-classified samples (p_t close to 1) and emphasizing samples with low confidence or high uncertainty. The parameter γ determines the strength of this focus: larger values make the loss concentrate more on hard or ambiguous samples. Finally, $\log(p_t)$ is the standard cross-entropy term that measures the discrepancy between predicted and true labels. In this study, because I-degree and IV-degree bamboo exhibit distinct phenotypic characteristics, they are easily classified and contribute less to the optimization process. Conversely, II-degree and III-degree bamboos display highly similar color-texture features, making them difficult to distinguish. By introducing the

weighting factor α_t and the focusing term $(1 - p_t)^\gamma$, the Focal Loss reduces the influence of easily classified samples while enhancing the learning of these transitional categories, thereby improving the model's ability to capture subtle phenotypic differences in Moso bamboo age classification.

The unsupervised loss function selects the Kullback-Leibler(KL) Divergence, which is a method for measuring the difference between two probability distributions. The specific formula(5) is:

$$L_{\text{unsup}} = \sum_x P(x) \log \left(\frac{P(x)}{Q(x)} \right) \quad (5)$$

where $P(x)$ is the predicted probability distribution of the teacher model, and $Q(x)$ is the predicted probability distribution of the student model. In this study, the KL divergence measures the consistency of the predictions of the unlabeled data between the student model and the teacher model, constrains the difference in the output distributions of the two models, and prevents the predictions of the student model from being over-confident or divergent, thereby leveraging the unlabeled data to enhance the generalization ability of the model. Additionally, by introducing the temperature parameter t , the effect of high-confidence pseudo-labels is further enhanced. The formula (6) is

$$P_{\text{teacher}}(c|x) = \frac{\exp(z_c/t)}{\sum_c \exp(z_c/t)} \quad (6)$$

where $P_{\text{teacher}}(c|x)$ denotes the probability that the teacher model assigns to class c for the input x , and z_c represents the logit value before the softmax operation. The temperature coefficient t modulates the sharpness of the distribution, when $t < 1$, the exponent term $\exp(z_c/t)$ enlarges the gap between logits, making the distribution more peaked. This emphasizes high-confidence pseudo-labels and suppresses low-confidence or ambiguous predictions. Conversely, larger values of t yield smoother probability distributions that increase tolerance to uncertainty.

Because transitional bamboo ages (e.g. II-degree and III-degree) exhibit subtle and overlapping phenotypic features, the use of a small temperature value ensures that the teacher model produces confident pseudo-labels for clearly distinguishable categories while maintaining stability in the pseudo-label refinement process. This enhances the consistency regularization between teacher and student models and ultimately improves the robustness of the semi-supervised learning framework.

Finally, the loss calculation formula (7) (8) of this study is as follows:

$$L = L_{\text{sup}} + \lambda(t) \cdot L_{\text{unsup}} \quad (7)$$

$$\lambda(t) = \begin{cases} 0, & e \leq 20 \\ \min(10, 5 * e/30), & e > 20 \end{cases} \quad (8)$$

In the formula, $\lambda(t)$ is the weight for the unsupervised loss, and e is the number of learning rounds. During the stage when the model is not yet stable and the quality of pseudo-labels is not high enough, the unsupervised loss may cause the model to deviate in learning the distribution. Therefore, the weight of the unsupervised loss is dynamically adjusted to avoid the interference of early pseudo-label noise. Meanwhile, the weight of the unsupervised loss needs to be continuously increased to improve the effectiveness of unlabeled samples in model training.

3.6. Age-dependent dynamic threshold mechanism

In semi-supervised learning, the quality of pseudo-labels is crucial for model performance. Unlike typical image classification models, the classification features of Moso bamboo age evolve with age, resulting in varying learning difficulties across different ages. Traditional fixed threshold strategies overlook this aspect, impairing model training and degrading pseudo-label quality and the effective use of unlabeled data.

To address this, we propose an adaptive dynamic threshold mechanism, The calculation process is as shown in the following:

$$\tau_t^{(k)} = \tau_0^{(k)} \cdot \exp(-\beta_t) - \delta \cdot \pi_{t-1}^{(k)} \quad (9)$$

Formula (9) defines the recursive update process of the threshold $\tau_t^{(k)}$ for category k at iteration t . Here, β_t represents the filtering weight that gradually decreases with training epochs. Because the early stage of training produces noisy pseudo-labels, a higher β_t value ensures stricter filtering. As training progresses and pseudo-labels become more reliable, β_t decays exponentially, allowing the model to incorporate more samples into learning. The parameter δ acts as a correction term to prevent the threshold from dropping excessively in consecutive iterations.

$$\tau_0^{(k)} = 0.95 - 0.05k, k = 1, 2, 3, 4 \quad (10)$$

Formula (10) defines the initial threshold $\tau_0^{(k)}$ for each bamboo age class. Because higher-degree bamboos are phenotypically more complex and harder to classify, they start with a slightly lower initial threshold to allow more samples to be included in the early pseudo-label pool.

$$\pi_{t-1}^{(k)} = \frac{P(k)}{\sum_{k=1}^K P(k)} \quad (11)$$

Formula (11) defines the normalized class probability proportion obtained from the previous training iteration. Here, $P(k)$ represents the total predicted probability of class k aggregated across all unlabeled samples, and $\sum_{k=1}^K P(k)$ is the sum of predicted probabilities over all classes. Therefore, $\pi_{t-1}^{(k)}$ reflects the relative contribution or dominance of each category in the model's predictions during the preceding iteration. Because a lower $\pi_{t-1}^{(k)}$ indicates that class $\pi_{t-1}^{(k)}$ was predicted less frequently or with lower confidence, this metric effectively measures the learning difficulty of each class. Consequently, the subsequent threshold $\tau_t^{(k)}$ is adaptively decreased for such underrepresented categories, allowing more pseudo-labels from these difficult classes to be incorporated into training and thereby maintaining inter-class balance in the semi-supervised process.

$$\delta = \begin{cases} 1, & \tau_{t-1}^{(k)} < 0.05 \\ 0, & \tau_{t-1}^{(k)} \geq 0.05 \end{cases} \quad (12)$$

Formula (12) defines the binary adjustment factor δ . When the proportion of pseudo-labels for a specific class falls below 5%, δ is activated to correct for under-representation by relaxing the threshold. When the class already has sufficient pseudo-labels, δ is set to zero to maintain stability.

In addition, pseudo-label selection must simultaneously satisfy two constraints: $\max(p) > \tau_t^{(k)}$, ensuring that only high-confidence predictions are retained; and $\max(p) - p_{top2} > 0.1$, filtering out samples with ambiguous inter-class probabilities. These dual conditions collectively guarantee that the pseudo-labels used for training are both confident and discriminative, which is essential for the phenotypically overlapping categories such as II-degree and III-degree bamboos.

3.7. Experimental conditions and evaluation indicators

The experimental environment of this study is shown in Table 2

Table 2
Experimental environment.

Environment	Configuration
platform	PyTorch
CPU	AMD Ryzen 7 5800H 8-core processor(3.20 GHz)
GPU	NVIDIA GeForce RTX 3060 Laptop 6GB
RAM	16GB
operating system	Windows11 64
software	CUDA11.1, CUDNN8.2.1, OpenCV4.5.5.64

below.

After the data in the dataset is segmented by SAM, data augmentation and training pre-processing are performed using the methods provided by albumentations [25]. Then, the augmented dataset is input into the model for training. The selected data augmentation methods are shown in the Table 3 below.

To ensure the robustness and reproducibility of the results, this study conducted five independent repetitions for all key experiments, each time using a different random seed for model initialization, data partitioning, and data augmentation. In each repetition, we trained the model with the same dataset split and recorded the final test set metrics. The mean $\pm$ standard deviation (std) is reported in the text and tables. To comprehensively evaluate the model's discrimination performance across different Moso bamboo age categories, we adopted four metrics: Test Accuracy, Macro Precision, Macro Recall, and Macro F1-score. These metrics give each category equal weight, thus avoiding bias caused by data imbalance.

Accuracy is used to measure the proportion of accurately predicted samples among all predicted samples, and the formula(13) is:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (13)$$

For each category i , the formula for Macro Precision is (14) (15):

$$precision_i = \frac{TP_i}{TP_i + FP_i} \quad (14)$$

$$Macro\ precision = \frac{1}{N} \sum_{i=1}^n precision_i \quad (15)$$

where TP_i represents the number of true positives for class i , and FP_i represents the number of false positives for class i . This metric reflects the model's average predictive precision across all categories and effectively measures the model's recognition quality for minority classes.

The formula for Macro Recall is (16) (17):

$$recall_i = \frac{TP_i}{TP_i + FN_i} \quad (16)$$

$$Macro\ recall_i = \frac{1}{N} \sum_{i=1}^n recall_i \quad (17)$$

Where FN_i represents the number of false negatives for class i . This metric measures the model's overall detection ability for all categories, with a particular focus on hard-to-recognize classes.

Macro F1 is a metric that balances Macro Precision and Macro Recall, with the formula (18)(19) is:

$$F1_i = \frac{2 * precision_i * recall_i}{precision_i + recall_i} \quad (18)$$

$$Macro\ F1 = \frac{1}{N} \sum_{i=1}^n F1_i \quad (19)$$

Table 3

Data augment method hyperparameter set.

Method	Parameter
Resize	240 * 240
CenterCrop	224 * 224
HorizontalFlip	P = 0.4
VerticalFlip	P = 0.4
ColorJitter	brightness=0.4, contrast=0.4, saturation=0.4, hue=0.1, p=0.8
GaussianBlur	blur_limit=(3, 7), sigma_limit=0.5, p=0.5
Normalize	mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]
ToTensorV2	

The hyperparameters used in training are shown in Table 4.

Table 4

Hyperparameter set.

Hyperparameter	set
Epochs	300
Batch Size	32
Learning rate	1e-2
Optimizer	AdamW
Weight decay	0.02
KL temperature T	0.7
β_t of $\tau_t^{(k)}$	$0.1 \cdot (1 + \text{epoch}/100)$
α_t of Focal Loss	0.25
γ of Focal Loss	0.1

4. Experimental result

4.1. Model performance evaluation

This study compared our proposed method with several advanced approaches, including ResNet18, ResNet-50 [26], Swin-Transformer [27], ShuffleNetV2 [28], MobileNetV3 [29], and the semi-supervised frameworks FixMatch and FlexMatch. The ResNet series, known for its powerful residual network architecture, remains one of the most outstanding models in the field of deep learning today. Swin-Transformer effectively combines the concepts of CNNs and Transformers, achieving state-of-the-art (SOTA) performance in many image processing domains. MobileNet and ShuffleNet are two efficient and powerful lightweight convolutional networks that significantly reduce network parameters while maintaining excellent feature extraction capabilities. We conducted five independent repetitions for the main comparative experiments, and the results are shown in Table 5.

The "Supervised" models were trained solely on the limited labeled training set (1,647 images). In contrast, the "SSL" models were designed to leverage both this small labeled set and the large pool of unlabeled data (5,396 images), representing a labeled-to-unlabeled data ratio of 1:3.3. This means that the SSL methods, including ours, were effectively learned from a scenario where only about 23% of the available training data was labeled. The superior performance of our DCT-MBADNet (SSL) under this setting underscores its primary advantage: effectively mitigating labeled data scarcity by capitalizing on unlabeled images to enhance model generalization, all within a lightweight architecture.

As shown in the table, the proposed DCT-MBADNet, when combined with the Mean Teacher framework and our age-dependent dynamic threshold mechanism, achieved the highest test set accuracy of 89.6%. Compared to ResNet-50, ShuffleNetV2, and MobileNetV3, our model surpassed them in accuracy, recall, and F1 scores for the Moso bamboo

age determination task, while simultaneously reducing the parameter count by 93%, 31%, and 70%, respectively. This demonstrates that the DCT-MBADNet network structure is lightweight yet highly accurate, meeting the demands for efficient and precise Moso bamboo age determination. Compared to the fully supervised version of DCT-MBADNet, the model under the semi-supervised paradigm improved test accuracy, macro precision, macro recall, and macro F1 by 4.5%, 4.8%, 5.0%, and 3.2%, respectively, validating the enhanced generalization ability gained from utilizing unlabeled data and the dynamic threshold mechanism.

Furthermore, in the fully supervised learning scenario, the performance of DCT-MBADNet is close to that of ResNet-50 and MobileNetV3, and significantly better than Swin-Transformer. This result indicates that while Swin-Transformer excels on large-scale datasets, its complex structure struggles to leverage its advantages in small-sample conditions and is constrained by data scarcity. In contrast, DCT-MBADNet, through dual-stream feature decoupling and dynamic feature focus, can effectively capture the color and texture phenotypic features of Moso bamboo and, to some extent, inhibit overfitting, thereby demonstrating superior stability and adaptability in scenarios with limited labeled data.

4.2. Ablation study

To validate the contribution of each module, this study designed ablation experiments, and the results are shown in Table 6, confusion matrices as shown in Fig. 7.

As shown in Table 6, each module played a key role in the improvement of overall performance. First, after removing the SAM mask segmentation, the accuracy dropped significantly from 89.6% to 71.2%, and the Macro F1 decreased from 88.3% to 65.9%. This indicates that complex backgrounds strongly interfere with feature extraction, and the introduction of the SAM mask enables the model to focus on the bamboo culm region, providing a more reliable input for subsequent

Table 6

Ablation study on model components (mean±std, n=5).

Variant	Test Acc (%)	Macro F1 (%)	F1(II) (%)	F1(III) (%)
w/o SAM mask	71.2±0.2	65.9±0.5	56.7±0.3	71.3±0.5
Only color stream	84.2±0.8	81.9±0.9	78.6±1.1	79.1±0.8
Only texture stream	77.8±0.3	75.3±0.6	64.4±0.7	71.7±0.5
Dual w/o gating fusion	86.3±0.4	84.1±0.5	77.8±0.8	80.3±0.7
Fixed threshold ($\tau=0.8$)	84.8±0.2	81.2±0.3	74.3±0.5	79.5±0.4
ours	89.6±0.5	88.3±0.4	82.1±0.7	86.3±0.6

Table 5

Performance comparison of Moso bamboo age discrimination under different models and training paradigms (mean±std, n=5).

Method	Backbone	Params (M)	FLOPS (G)	Test Acc (%)	Macro F1 (%)	Macro Precision (%)	Macro Recall (%)
ResNet-50 (Supervised)	ResNet-50	23.5	4.09	86.1±0.8	84.6±0.9	85.7±0.7	83.1±0.9
Swin Transformer (Supervised)	Swin Transformer	50.3	8.7	78.9±1.1	73.4±1.3	75.6±0.9	71.7±1.1
ShuffleNetV2 (Supervised)	ShuffleNetV2	2.3	0.15	83.2±0.6	81.7±0.6	82.3±0.7	81.2±0.8
MobileNetV3 (Supervised)	MobileNetV3	5.4	0.23	85.8±0.5	83.2±0.5	82.5±0.6	80.9±0.8
DCT-MBADNet (Supervised)	ours	1.6	0.22	85.1±0.6	83.5±0.6	84.1±0.5	80.2±0.7
FixMatch (SSL)	ResNet-18	11.3	3.41	86.3±0.5	84.3±0.4	85.4±0.3	82.5±0.4
FlexMatch (SSL)	ResNet-18	11.3	3.61	87.2±0.3	85.1±0.4	88.2±0.4	80.3±0.5
DCT-MBADNet (SSL)	ours	1.6	0.22	89.6±0.5	88.3±0.4	89.1±0.4	83.4±0.3

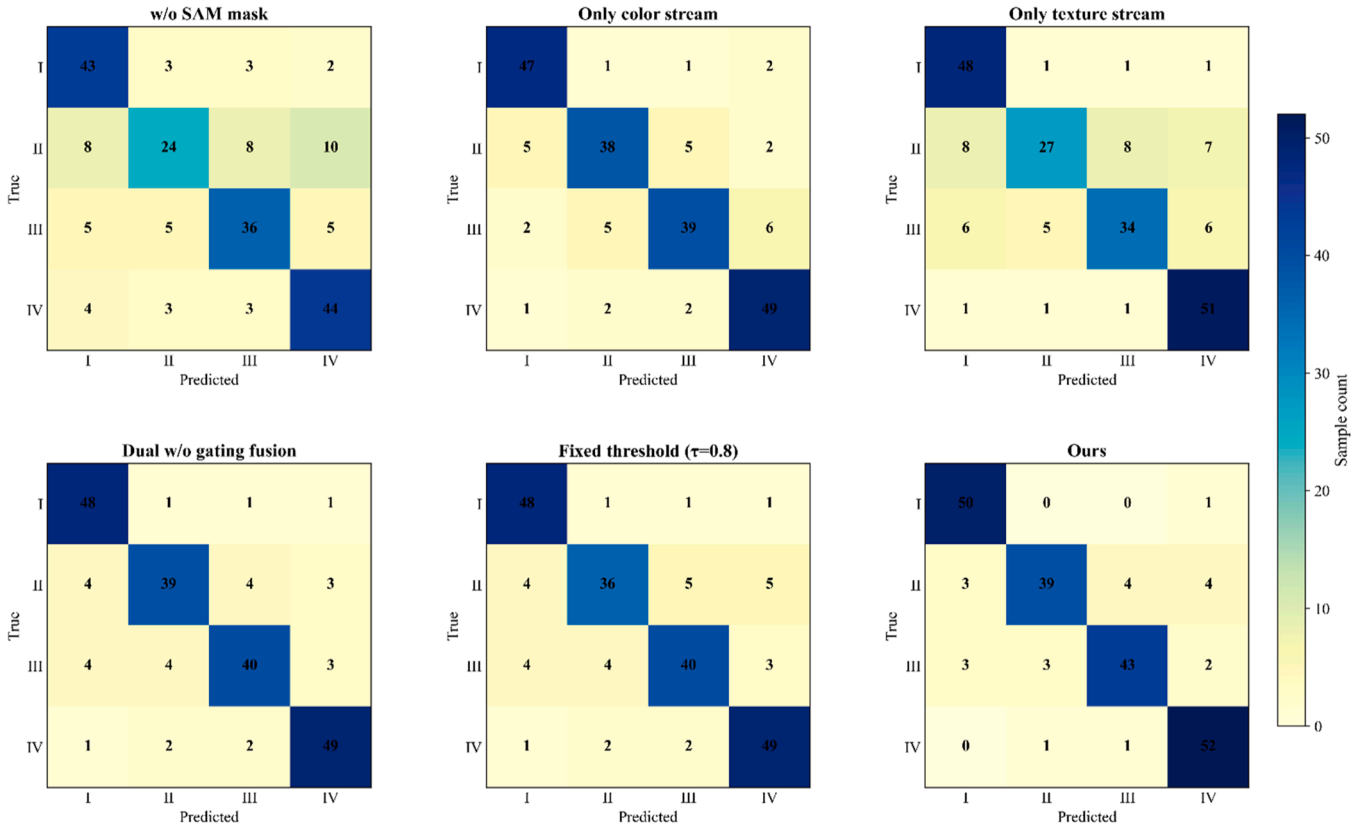


Fig. 7. Confusion matrices under different ablation experiments.

dual-stream decoupling.

In the single-stream experiments, using only the color stream or the texture stream resulted in accuracies of 84.2% and 77.8%, respectively. In comparison, the color stream shows a greater advantage in capturing the color degradation patterns of the culm, while the texture stream better reflects the roughening trend of the culm surface. However, using either stream alone is insufficient to capture the full range of Moso bamboo age features. This also indicates that color features are more important for age determination. Upon further fusion of the two, without introducing the gating mechanism, the model's accuracy improved to 86.3%, with a Macro F1 of 84.1%, demonstrating the complementarity of color and texture information in bamboo age determination.

Regarding pseudo-label filtering, the model's accuracy was 84.8% with a fixed threshold ($\tau=0.8$), and the Macro F1 was 81.2%, which is lower than that of the dynamic threshold strategy (89.6%). A fixed threshold that is too low can lead to high pseudo-label noise, while a threshold that is too high can result in too few unlabeled samples being utilized. In contrast, the age-dependent dynamic threshold mechanism proposed in this paper can adaptively adjust to the difficulty of each category and the training progress, prioritizing the use of high-confidence samples and gradually introducing difficult samples. This achieves an effective balance between the quality and quantity of pseudo-labels.

Furthermore, the performance improvement in transitional categories provides intuitive proof of this mechanism's effectiveness: under the fixed threshold strategy, the F1-scores for II-degree and III-degree bamboo were only 74.3% and 79.5%, respectively, while they increased to 82.1% and 86.3% under the dynamic threshold strategy. This fully demonstrates that the dynamic threshold can effectively alleviate the problem of transitional categories being difficult for the model to distinguish due to ambiguous phenotypic features.

In summary, the ablation experiments clearly show that the SAM

mask ensures the purity of the input, the dual-stream decoupling enables complementary modeling of color and texture, and the age-dependent dynamic threshold mechanism significantly improves the discriminative ability for transitional categories. The combined effect of these modules ultimately allows DCT-MBADNet to achieve better classification performance while remaining lightweight.

4.3. Visualization of grad-CAM feature heatmap

Grad-CAM (Gradient-weighted Class Activation Mapping) [30] is a visualization technique used to explain the decision-making process of deep convolutional neural networks. It determines which regions in an image contribute the most to the prediction of a specific class by computing the gradients of the target class with respect to the feature map of the last convolutional layer. In this study, we obtained the gradients by registering hooks to the last convolutional layers of the color and texture streams. The results for the four categories are shown in the Fig. 8.

From the results, it can be observed that the color stream heatmaps for all four categories cover the surface of the bamboo culm, proving that the color stream effectively extracts the color features of the culm as a basis for model prediction. Among these, the response for II-degree bamboo is the least intense, indicating that color information contributes minimally to its prediction. This is because the features of II-degree bamboo are the most transitional, with colors between I-degree and III-degree bamboo, making color information difficult to serve as a decisive criterion. This also aligns with the results from the confusion matrix. In contrast, the color stream response for I-degree bamboo is strong, suggesting that color is a crucial criterion for determining if Moso bamboo is I-degree. The texture stream heatmaps are more concentrated on the bamboo culm edges, internodes, sheath scars, and complex textures on the culm surface. The heat map areas for I and II-degree bamboo are larger and smoother, because the surfaces of I and II-degree bamboo are

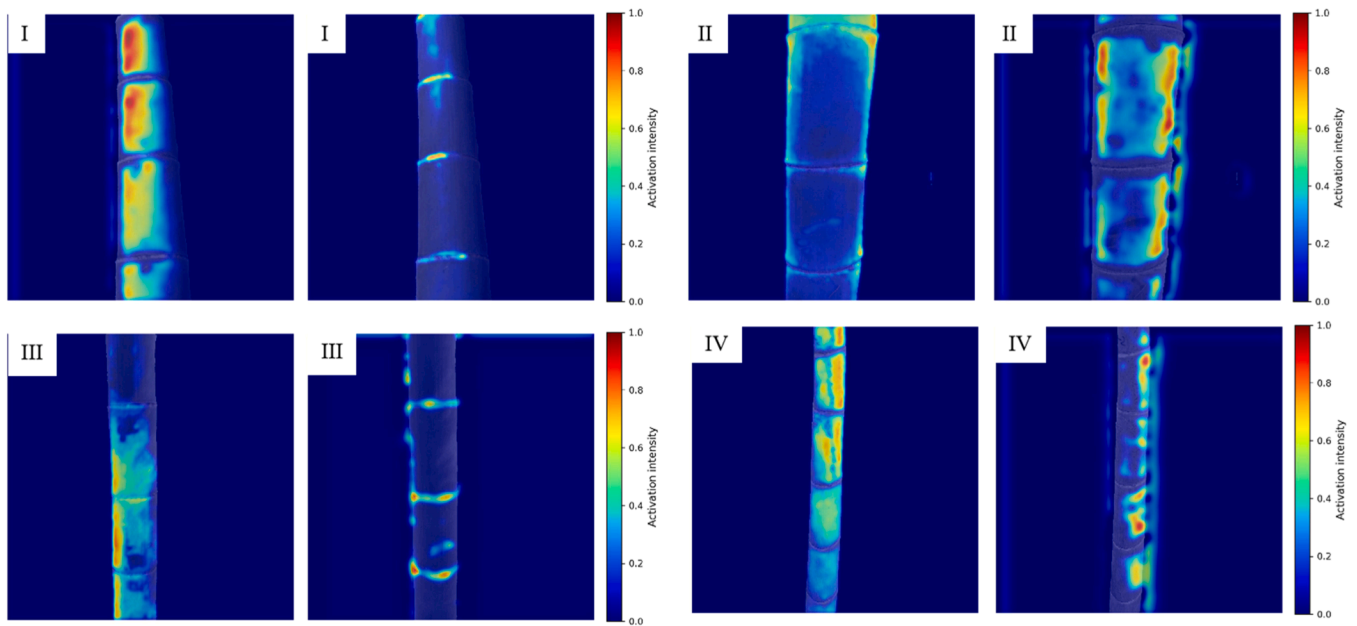


Fig. 8. Grad-CAM visualization of feature responses for different bamboo age classes. The color scale represents the activation intensity of the model, where red and yellow indicate regions with high feature importance contributing most to the age prediction, and blue denotes low-attention areas. Left: color stream; right: texture stream.

smoother and have fewer textures. Conversely, the texture stream heatmaps for III and IV-degree bamboo are more concentrated on the sheath scars and surface textures, which is consistent with the growth characteristics of Moso bamboo: as age increases, the sheath scars turn black and become rougher, and the culm surface develops denser textures. This proves the effectiveness of the texture stream's design in focusing on multi-scale texture features to provide a basis for the model's classification, and also reflects the efficacy of the dual-stream feature decoupling network.

5. Discussion

5.1. Analysis of core modules

The improvements achieved by DCT-MBADNet originate from how its architecture aligns with the biological and visual characteristics of Moso bamboo. The dual-stream structure is inspired by the fact that bamboo aging is jointly reflected in color fading and surface textural coarsening. By processing these two modalities separately, the color stream captures the gradual green-to-yellow spectral shift, while the texture stream learns multi-scale roughness patterns such as sheath-scar darkening and epidermal cracking. Their adaptive fusion dynamically balances these complementary cues and prevents either from dominating when its evidence is weak, particularly in transitional stages where phenotypic ambiguity is high. The ablation study shows that the SAM mask module plays a crucial role in the preprocessing stage. Without this module, the accuracy drops to 71.2% and the Macro F1 is only 65.9%. This strongly demonstrates that complex backgrounds can severely interfere with the model's feature extraction and confirms that background removal is a fundamental and critical task for improving the accuracy of forestry image analysis, a finding consistent with the conclusions of Carpentier et al. regarding the impact of background interference on tree bark recognition accuracy [31]. In addition, using a fixed threshold strategy leads to a significant performance decline. A threshold that is too low can introduce a high proportion of pseudo-label noise, while one that is too high filters out too many samples, reducing the utilization of unlabeled data. In contrast, the age-dependent dynamic threshold mechanism proposed in this paper can adaptively adjust the pseudo-label filtering criteria based on category difficulty and

training progress, thereby balancing the quality and quantity of pseudo-labels. For the transitional II and III-degree bamboo categories, this mechanism significantly alleviates the confusion caused by their transitional phenotypic features, demonstrating strong adaptability.

DCT-MBADNet, with only 1.6 M parameters, achieves an accuracy of 89.6%, surpassing ResNet-50 and MobileNetV3 and performing significantly better than Swin-Transformer. This suggests that in small-sample scenarios, the Transformer architecture struggles to fully leverage the advantages of its global attention mechanism. This finding is highly consistent with the known limitations of Transformer architectures in low-data regimes and the feature separation capabilities of convolutional networks [32]. Conversely, ResNet, ShuffleNet, and MobileNet, which use residual structures and are combined with task-specific modules, can effectively capture fine-grained phenotypic features related to bamboo age and, to some extent, suppress overfitting, thus exhibiting better stability and adaptability in limited-labeled scenarios. This also indicates that while complex information may require deeper network structures and larger parameter counts to separate and extract, this phenomenon can be greatly mitigated by using specific modules to focus on their respective tasks. Therefore, this study combines a residual network structure with a specialized dual-stream feature decoupling module to achieve better results in the field of Moso bamboo age determination.

Previous studies have shown that color and texture features are complementary in classification and recognition [33]. Juola et al. improved tree species identification accuracy by fusing spectral color and texture information [34], while Ratajczak et al. validated the effectiveness of a color-texture separation strategy for bark classification [35]. The dual-stream decoupling method proposed in this study not only extends this idea by precisely modeling the biological patterns of Moso bamboo's color degradation and texture roughening but also further incorporates a dynamic gating mechanism to overcome the limitations of single-stream or simple concatenation methods. At the same time, the age-dependent dynamic threshold mechanism introduces biological prior knowledge, which is a factor not considered by existing semi-supervised methods, effectively solving the confusion problem in transitional categories.

The lightweight design of DCT-MBADNet makes it suitable for deployment on mobile devices, supporting the real-time detection needs

for bamboo forest resource surveys and carbon sink measurement. By synergistically using a small amount of labeled data and a large volume of unlabeled data, this research provides a low-cost, robust solution for forestry parameter extraction, aligning with the research philosophy of other advanced forestry methods [36–38].

5.2. Limitation and future work

Despite its good performance, DCT-MBADNet still has some limitations. Although DCT-MBADNet itself is a lightweight model, its preprocessing step relies on SAM, a model with a huge number of parameters. Therefore, in both this study and practical applications, SAM is more suitable as an offline preprocessing tool rather than being integrated into a real-time inference system, which ensures the advantages of its lightweight network are fully realized in field deployment scenarios.

Secondly, the feature confusion problem in transitional categories persists most of the model's prediction errors are concentrated on misclassifications between adjacent II-degree and III-degree bamboo, primarily because the features of these two categories are highly transitional, with subtle phenotypic differences that remain difficult to distinguish. Future work can be improved in the following aspects: (1) More multi-dimensional feature extraction: Expand the current number of channels and explore extracting richer color statistical features in different color spaces such as YUV and Lab to quantify the color differences more comprehensively in Moso bamboo. (2) Loss function optimization: Explore the introduction of more advanced loss functions, such as Triplet Loss and Contrastive Loss, to increase inter-class separation and reduce intra-class variation, thereby enhancing the model's discriminative ability for adjacent categories. (3) Image brightness optimization: Optimize images under different lighting conditions based on the Retinex theory and other advanced methods to make color representation more consistent. (4) Increased model complexity: Introduce more diverse attention mechanisms to focus on subtle changes in the texture of the bamboo internodes.

An important limitation of this study lies in its seasonal and geographical scope. All image data were acquired during the summer months (July–August), a period chosen for the relative stability and distinctiveness of Moso bamboo's color and texture traits—key features underpinning our model. However, this seasonal focus inherently constrains the model's applicability across other times of the year. Phenotypic variations induced by winter weathering, moisture fluctuation, or epiphytic growth (e.g., moss or lichen) remain unaddressed. Future work should therefore prioritize the construction of a multi-seasonal dataset, capturing the same bamboo stands across spring, autumn, and winter, to better understand phenotypic dynamics and ultimately develop a season-invariant age determination model.

Similarly, the model's regional generalizability requires further validation. Our dataset was drawn primarily from three geographical units within Fujian Province. Variations in climate, soil, and Moso bamboo subspecies across broader regions—such as the Yunnan–Guizhou Plateau and Jiangsu–Zhejiang area—may lead to divergent growth patterns and phenotypic expressions. To improve robustness and generalization, we plan to compile a more extensive and geographically diverse image corpus, incorporating multi-season and multi-lighting condition samples, supported by shoot-emergence tagging for accurate age grounding.

Although this study has achieved excellent experimental results, the mathematical formulas of the dynamic threshold mechanism contain several empirical parameters, and their convergence and optimality still require more rigorous theoretical support. Future work will attempt to provide a more in-depth theoretical analysis, for example, by conducting a sensitivity analysis to explore the impact of parameter selection on model performance and investigating the conditions under which this dynamic strategy can ensure a continuous improvement in pseudo-label quality.

6. Conclusion

This study addresses the problems of scarce labeled data, complex background interference, and classification ambiguity caused by gradual phenotypic features in Moso bamboo age determination. We propose a semi-supervised framework based on color-texture decoupling, DCT-MBADNet. Through systematic experiments and theoretical analysis, we draw the following conclusions:

- (1) The SAM-based bamboo culm segmentation module effectively eliminates background noise, improves model accuracy, and lays the foundation for independent modeling of color degradation and texture evolution features by the color and texture streams. The dual-stream feature decoupling also validates the strong correlation between phenotypic features and age.
- (2) The proposed DCT-MBADNet network architecture is well-designed and performs exceptionally well. Under the fully supervised learning paradigm, it achieves accuracy comparable to that of ResNet-50, ShuffleNetV2 and MobileNetV3 while reducing the parameter count by 93%, 31% and 70%, respectively. The core dual-stream decoupling encoding module effectively extracts the color and texture features of Moso bamboo. Under the semi-supervised learning paradigm, it achieves the highest test accuracy, macro precision, Macro recall, and macro F1, at 89.6%, 89.1%, 83.4%, and 88.3%, respectively. This not only further illustrates the advantages of the DCT-MBADNet network structure but also confirms that the semi-supervised learning paradigm enhances the model's generalization ability. At the practical application level, DCT-MBADNet, by synergistically utilizing a small amount of labeled data and a large volume of unlabeled data, provides a low-cost, highly robust technical solution for precise bamboo forest management. Its adaptability to handheld devices for grassroots foresters makes it suitable for further promotion in scenarios such as bamboo forest resource surveys, felling planning, and carbon sink measurement, assisting the collaborative advancement of forestry digitalization and rural revitalization strategies.
- (3) This study proposes a dynamic threshold mechanism for Moso bamboo age determination under a semi-supervised learning paradigm, which uses a curriculum learning strategy to adaptively filter high-confidence pseudo-labels. Experiments show that after introducing the age-dependent dynamic threshold mechanism, the F1-scores for II and III-degree bamboo improved to 82.1% and 86.3%, respectively, compared to the fixed-threshold mechanism. This mitigates the problems of low-quality pseudo-labels or the inability to effectively learn from unlabeled samples, as well as the classification ambiguity caused by the highly transitional nature of Moso bamboo's category features.
- (4) Although DCT-MBADNet performs excellently, some issues remain: firstly, II and III-degree bamboo are still confused due to similar phenotypic features. In the future, more advanced techniques can be introduced to improve feature extraction and further distinguish transitional categories. Secondly, the current model's generalization ability is limited by the dataset size. Future work will build a larger-scale dataset to enhance the model's generalization.

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Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Ethical statement

I certify that this manuscript is original and has not been published and will not be submitted elsewhere for publication while being considered by Smart Agriculture Technology. And the study is not split up into several parts to increase the quantity of submissions and submitted to various journals or to one journal over time. No data have been fabricated or manipulated (including images) to support your conclusions. No data, text, or theories by others are presented as if they were our own.

The submission has been received explicitly from all co-authors. And authors whose names appear on the submission have contributed sufficiently to the scientific work and therefore share collective responsibility and accountability for the results."

CRediT authorship contribution statement

Zhihui Yu: Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation. **Hanyue Song:** Writing – review & editing, Supervision, Investigation. **Xiang Huang:** Writing – review & editing. **Yangyang Zhang:** Investigation. **Mingxin Li:** Investigation. **Simei Lin:** Writing – review & editing. **Jian Liu:** Validation. **Kunyang Yu:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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